

AIDEN COX

Dayton, Ohio

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Education

Wright State University

August 2023 – May 2026

Bachelor of Science in Computer Science | GPA: 4.0 / 4.0

Fairborn, Ohio

Relevant Coursework

- Data Structures
- Intro To Machine Learning
- Intro To Cyber Security
- Mobile Development
- Intro to Software Engineering
- Host Computer Security

Experience

Wright-Patt Credit Union

May 2025 – Present

Year-round intern - Data Analytics / Automation

Fairborn, Ohio

- Supported business operations by streamlining data workflows and automating repetitive tasks in Excel using VBA, resulting in increased efficiency and reduced errors. Developed and maintained interactive dashboards in PowerBI to visualize key performance indicators, trends, and actionable insights. Collaborated with team members to gather requirements, clean datasets, and deliver analytics solutions that informed strategic decisions.
- Using Excel, Visual Studio Code, Python, Jupyter Notebook, and PowerBI to complete tasks.

Teaching Assistant

August 2024 – Present

Operating Systems, Concepts, and Usage - Lab Leader

Fairborn, Ohio

- Lead a Lab of 40-50 students teaching core concepts of Computer Science. Duties included assisting students during the lab session by giving students one-on-one attention. Help grade and provide feedback on lab assignments. Holding two in-person office hours a week, and aided in monitoring questions and conversations in the Discord channel.
- Students used Visual Studio Code, Terminal, WSL2 + Ubuntu, Bash, and Git.

Projects

JPL Construction App - BufferFlushed Labs LLC | Expo, Python, Firestore, TypeScript

May 2025 - Current

- Collaborated cross-functionally to develop a scalable, mobile-responsive Job Finance App aimed at streamlining business operations for invoicing, client tracking, and financial reporting.
- Engineered dynamic data models for job, client, and invoice management using real-time cloud storage and integrated automated PDF generation and email dispatch functionality.
- Designed and implemented a secure customer portal to manage payments and client interactions, improving user experience and reducing administrative workload.
- Integrated planned enhancements including data-driven financial dashboards, automated recurring billing, and animated UI transitions to boost engagement and usability.
- Utilized Agile development practices, version control, and team-based planning tools to ensure efficient project delivery and high-quality codebase maintenance.

Hash Table | C++, CLion, Valgrind

March 2025

- Engineered a custom hash table data structure in C++, implementing separate chaining to resolve collisions and ensure consistent performance under high load.
- Integrated dynamic resizing to maintain optimal load factors, automatically rehashing and distributing entries to preserve $O(1)$ amortized access times.
- Designed and implemented a custom hash function to reduce clustering and evenly distribute keys, improving lookup performance and reducing collision rates.
- Focused on memory efficiency by dynamically allocating node chains and managing de-allocation to prevent memory leaks.
- Validated performance through unit testing and benchmarking against standard C++ map structures.

Undergraduate AI Research with Dean | Python Orange

May 2024

- Comprehensive approach to testing artificial intelligence, incorporating elements such as data verification, handling missing data, addressing the decision problem and meticulous experimental design.

Leadership / Extracurricular

- * Tau Beta Pi Engineering Honor Society Treasurer
- * ACM Programming Contest 2024 contestant and 2025 Judge.
- * Computer Engineering and Computer Science Alumni Society Board Member